

ELECTRIFICATION OF BUSES USING ÚJPEST-KÖZPONT TERMINAL

- Recep Eren Durgut | C1FYSH
- Joywin Jebii Tomno | JB8W27
- Daisy Machanja | FZIZ3Y
- Mateo Andrade | YS4HDL

Budapest University of Technology and Economics
Faculty of Transportation Engineering and Vehicle Engineering – Department of Transport Technology and Economics

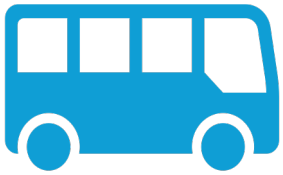
INTRODUCTION



Goal

Replacing ICE buses with electric buses on lines

25, 147, 170, 270, 220.



Methodology

Iterative heuristic

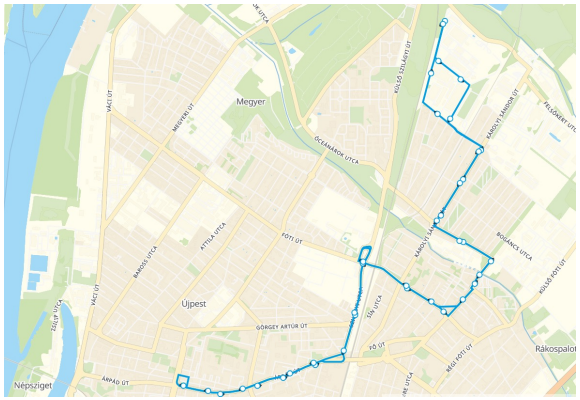


Vehicle Type

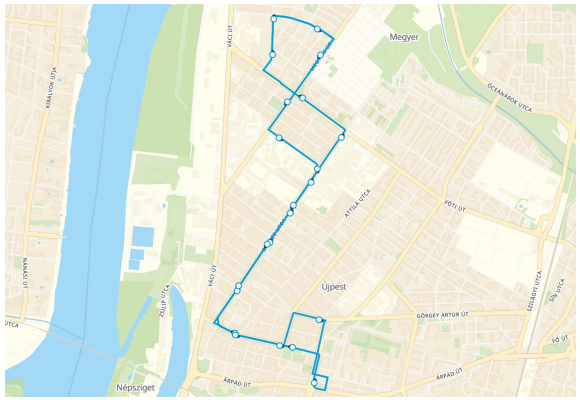
Solo buses

INTRODUCTION

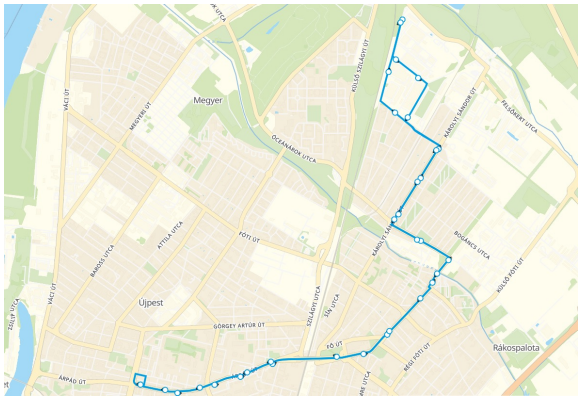
Line 270



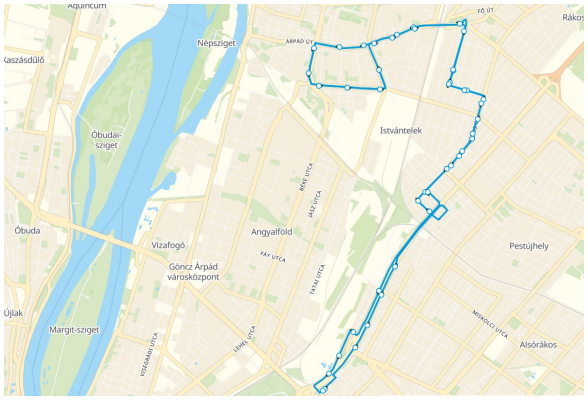
Line 147



Line 170



Line 25



Line 220



Source: https://futar.bkk.hu/route/

METHODOLOGY

STEP 1

Linear Programming on Python

Pros: Automatic calculations, scaleble, low code required.

Cons: Hard to implement time constraints, becomes NP-hard when break, driver change and charging introduced

METHODOLOGY

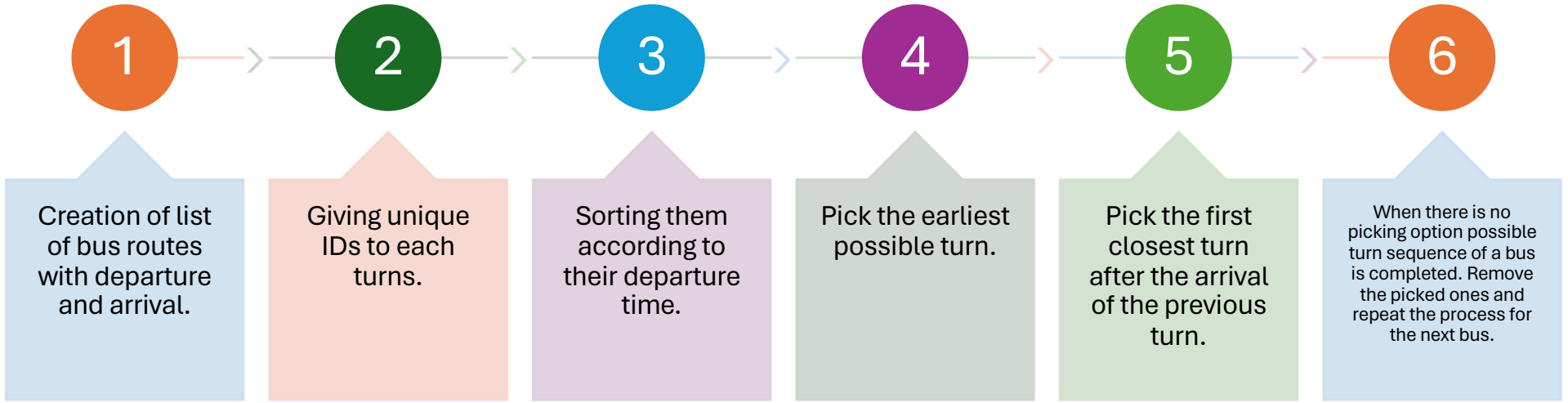
STEP 2

Manual Iterative Heuristic on Excel

Pros: Easy to implement, no code required, adaptable

Cons: Manual work, not scalable, time consuming, issues requires complete redo

TURN SEQUENCE APPROACH



TURN SEQUENCE APPROACH

Arrival	ID	Line	Departure	Arrival	Time in Hour	Taken?	KM/H	Consumption
04:27	E1	220	04:09	04:27	0,3	1	33,6	2,6
04:46	A0	1470	04:23	04:46	0,38333333	0	27,1304348	2,6
04:47	E28	220	04:29	04:47	0,3	1	33,6	2,6
05:07	E2	220	04:49	05:07	0,3	0	33,6	2,6
05:17	A1	1470	04:53	05:17	0,4	0	26	2,6
05:28	E29	220	05:09	05:28	0,31666667	1	31,8315789	2,6
05:35	A2	1470	05:10	05:35	0,41666667	0	24,96	2,8
05:59	D0	25	05:12	05:59	0,78333333	0	21,8680851	2,8
05:53	B0	1700	05:16	05:53	0,61666667	0	21,4702703	2,8
05:40	E3	220	05:21	05:40	0,31666667	0	31,8315789	2,6
05:55	A3	1470	05:30	05:55	0,41666667	1	24,96	2,8
06:19	D1	25	05:32	06:19	0,78333333	0	21,8680851	2,8
05:52	E30	220	05:33	05:52	0,31666667	0	31,8315789	2,6
06:13	B1	1700	05:36	06:13	0,61666667	0	21,4702703	2,8

ID	Line	Turn Sequence	Consumption	Remaning Energy	t	t(0)	Charging t	Driver t
E1	220	1	26,208	117,792	4:27	00:18	0	00:18
E28	220	1	26,208	91,584	4:47	00:38	0	00:38
Charge		1			4:58	00:49	0,19648	00:49
E29	220	1	26,208	153,792	5:28	01:19	0	01:19
A3	1470	1	29,12	124,672	5:55	01:46	0	01:46
B2	1700	1	37,072	87,6	6:31	02:22	0	02:22
Charge		1			6:43	02:34	0,20533333	02:34
A8	1470	1	29,12	150,88	7:12	03:03	0	03:03
E60	220	1	26,208	124,672	7:36	03:27	0	03:27
Charge		1			7:43	03:34	0,12295111	03:34
A13	1470	1	29,12	150,88	8:14	04:05	0	04:05
D20	25	1	53,103	97,777	9:15	05:06	0	05:06
Break		1		97,777	10:00	05:51	0,75	05:51
Charge		1			10:10	06:01	0,18271778	06:01

Source: https://bmeedu-my.sharepoint.com/:x:/g/personal/receperen_durgut_edu_bme_hu/EWYHYb_ca3tDgQTqCQU1BIMBe-MzhAhey_5PsUkrL8MYEQ?e=MWh64R

TURN SEQUENCE APPROACH

Charging Phase ID	Start of Charge	End of Charge	Duration	Charger	
1	4:47	4:58	0:11	Charger 1	
11	5:07	5:18	0:11	Charger 1	
21	5:17	5:25	0:08		0
52	5:52	6:00	0:08	Charger 1	
43	5:53	6:02	0:09		0
30	6:13	6:26	0:13	Charger 1	
81	6:21	6:29	0:08		0
2	6:31	6:43	0:12	Charger 1	
12	6:32	6:41	0:09		0
66	6:34	6:45	0:11		0
59	6:39	6:51	0:12		0
74	6:45	6:56	0:11	Charger 1	
22	6:47	6:57	0:10		0
108	7:02	7:10	0:08	Charger 1	
88	7:15	7:26	0:11	Charger 1	
94	7:23	7:34	0:11		0

ID	t	t(0)		Charger
1	4:58		00:11	Charger 1
11	5:18		00:23	Charger 1
52	6:00		00:31	Charger 1
30	6:26		00:45	Charger 1
2	6:43		00:57	Charger 1
74	6:56		01:09	Charger 1
108	7:10		01:17	Charger 1
88	7:26		01:29	Charger 1
100	7:44		01:43	Charger 1
31	8:06		01:53	Charger 1

TURN SEQUENCE RESULTS



CURRENTLY

THERE ARE 21 BUSES
ON THE ROUTES.



ELECTRIFIED

WITH 22 BUSES ALL
TURNS CAN BE DONE.

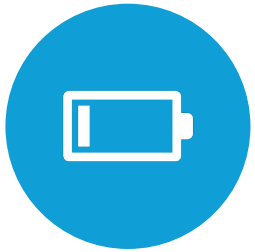
CHARGING RULES



BATTERY STATE OF CHARGE CANNOT BE BELOW 40%
(=> 72 KW)



BUS WILL CHARGE UNTIL ITS SOC IS 100%
(<=180 KW)



BUS WILL START ITS TURN WITH 80% CHARGE
(ACCOUNTS FOR DEPOT TURN)



RATED POWER OF A TERMINAL CHARGER IS 450 KW

CHARGING LOCATION: Petofi Utca and Kassai Utca



CHARGING LOCATION: Petofi Utca and Kassai Utca

- 5 chargers required.
- Bus length: 12 meters
- 5 charging buses: 60 meters
- Maneuver area required.
- 3 meters gap: 12 meters
- Total length required: 72 meters

RESULTS

Bus Utilization	Bus 1	Bus 2	Bus 3	Bus 4
Working Time	19:18	19:19	18:18	17:01
On Route Time	14:19	13:58	13:23	12:54
	74%	72%	73%	76%

Charger Utilization	Charger 1	Charger 2	Charger 3	Charger 4
Total Occupation Time	18:02	16:59	14:49	10:53
Real Charging Time	10:33	07:08	03:54	01:44
	59%	42%	26%	16%

POWER GENERATION AND ENERGY STORAGE

SOLAR RADIATION MAP - ÚJPEST-KÖZPONT TERMINAL



Source: https://www.mnnsz.hu/szolarterkep/

CHARGING TIMETABLE

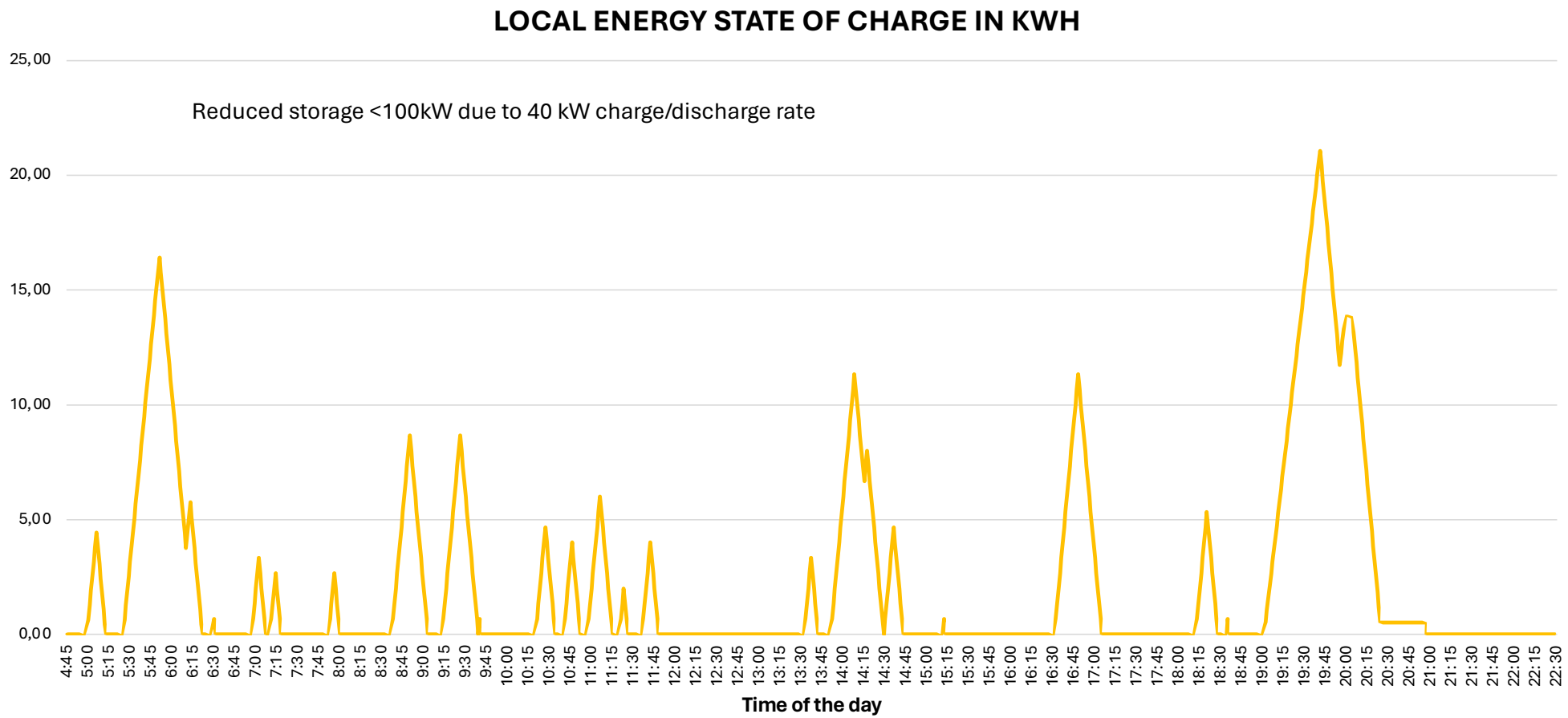
Start of Charge	Duration	End of Charge	Turn	Charger
4:47	0:11	4:58	1	Charger 1
5:07	0:11	5:18	2	Charger 1
5:17	0:08	5:25	3	Charger 2
5:52	0:08	6:00	7	Charger 2
5:53	0:09	6:02	6	Charger 2
5:59	0:11	6:10	5	Charger 1
6:13	0:13	6:26	4	Charger 1
6:21	0:08	6:29	11	Charger 1
6:31	0:12	6:43	1	Charger 1
6:32	0:09	6:41	2	Charger 2
6:34	0:11	6:45	9	Charger 3
6:39	0:12	6:51	8	Charger 1
6:45	0:11	6:56	10	Charger 4
6:47	0:10	6:57	3	Charger 2
7:02	0:08	7:10	16	Charger 3
7:15	0:11	7:26	12	Charger 2
7:23	0:11	7:34	13	Charger 1
7:26	0:09	7:35	5	Charger 1
7:31	0:13	7:44	14	Charger 4
7:36	0:07	7:43	1	Charger 2
7:38	0:11	7:49	6	Charger 2
7:41	0:11	7:52	15	Charger 1
7:56	0:10	8:06	4	Charger 1
7:58	0:10	8:08	11	Charger 3
8:02	0:14	8:16	17	Charger 3

POWER GENERATION AND ENERGY STORAGE

Roof	Solar power output	Suitable Roof Area	Total Roof Area
	kW	m^2	m^2
Metro Roof	10.8	49	129
Jozsef Attila	31.2	140	442
In-between	1.2	5	20
Petofi Utca	20	90	278
	63.2	**284**	**869**

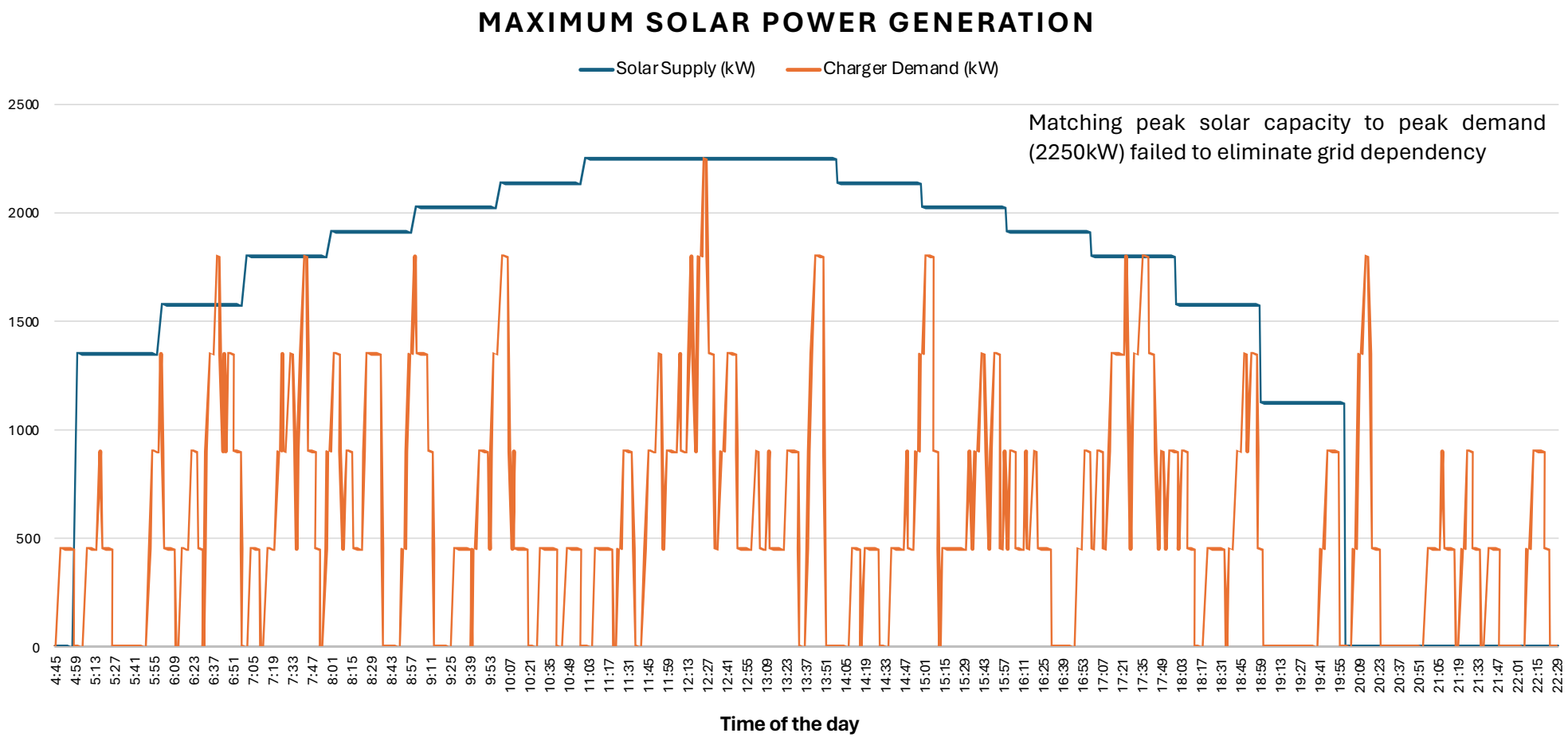
Time	Solar Supply (kW)	Charger Demand (kW)	Net Power (kW)	Battery Status (kWh)	Grid Intake (kW)
4:55	0.0	450	-450.00	0.00	450.00
4:56	0.0	450	-450.00	0.00	450.00
4:57	0.0	450	-450.00	0.00	450.00
4:58	0.0	450	-450.00	0.00	450.00
4:59	0.0	0	0.00	0.00	0.00
5:00	37.9	0	37.92	0.63	0.00
5:01	37.9	0	37.92	1.26	0.00
5:02	37.9	0	37.92	1.90	0.00
5:03	37.9	0	37.92	2.53	0.00
5:04	37.9	0	37.92	3.16	0.00
5:05	37.9	0	37.92	3.79	0.00
5:06	37.9	0	37.92	4.42	0.00
5:07	37.9	450	-412.08	3.76	372.08
5:08	37.9	450	-412.08	3.09	372.08
5:09	37.9	450	-412.08	2.42	372.08
5:10	37.9	450	-412.08	1.76	372.08
5:11	37.9	450	-412.08	1.09	372.08
5:12	37.9	450	-412.08	0.42	372.08
5:13	37.9	450	-412.08	0.00	386.64
5:14	37.9	450	-412.08	0.00	412.08

POWER GENERATION AND ENERGY STORAGE



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POWER GENERATION AND ENERGY STORAGE



Source: https://bmeedu-my.sharepoint.com/:x:/g/personal/receperen_durgut_edu_bme_hu/EWYHYb_ca3tDgQTqCQU1BIMBe-MzhAhey_5PsUkrL8MYEQ?e=MWh64R

CONCLUSION : PROS



METHODOLOGY IS FOOLPROOF



EASY TO UNDERSTAND

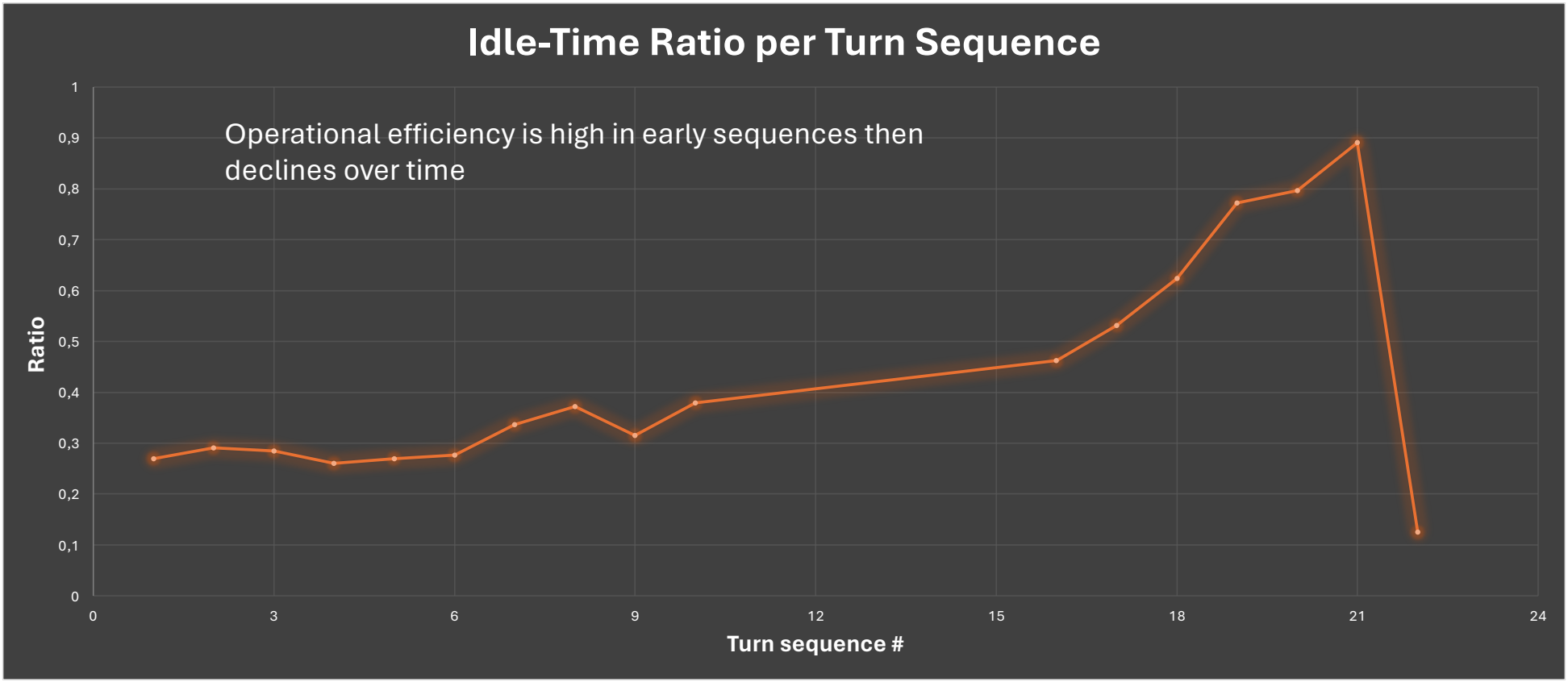


EASY TO IMPLEMENT

CONCLUSION : CONS

- Very repetetive.
- Easy to make mistakes.
- Mistakes are hard to detect and when they happen, hard to correct.
- Dwelling times increase overtime as methodology tries to moves iteratively. It doesn’t consider changing the previous turn sequences.

CONCLUSION : CONS



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QUESTIONS!